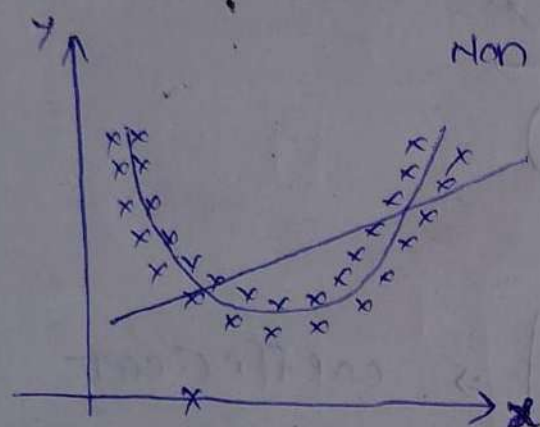


Polynomial Regression

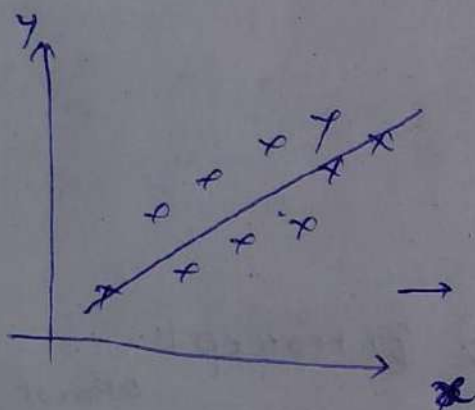


Non linear Relationship

$$h_0(x) = \beta_0 + \beta_1 x$$

- simple linear Regression

Error ↑↑



→ Hyperplane

Error ↓↓ → Polynomial Regression

Polynomial Degrees

degree = 0

simple linear Regression

degree = 0

Polynomial degree = 0

$$h_0(x) = \beta_0 x^0 \Rightarrow \text{Constant value}$$

1 i/p and 1 o/p feature

Polynomial
degree = 1

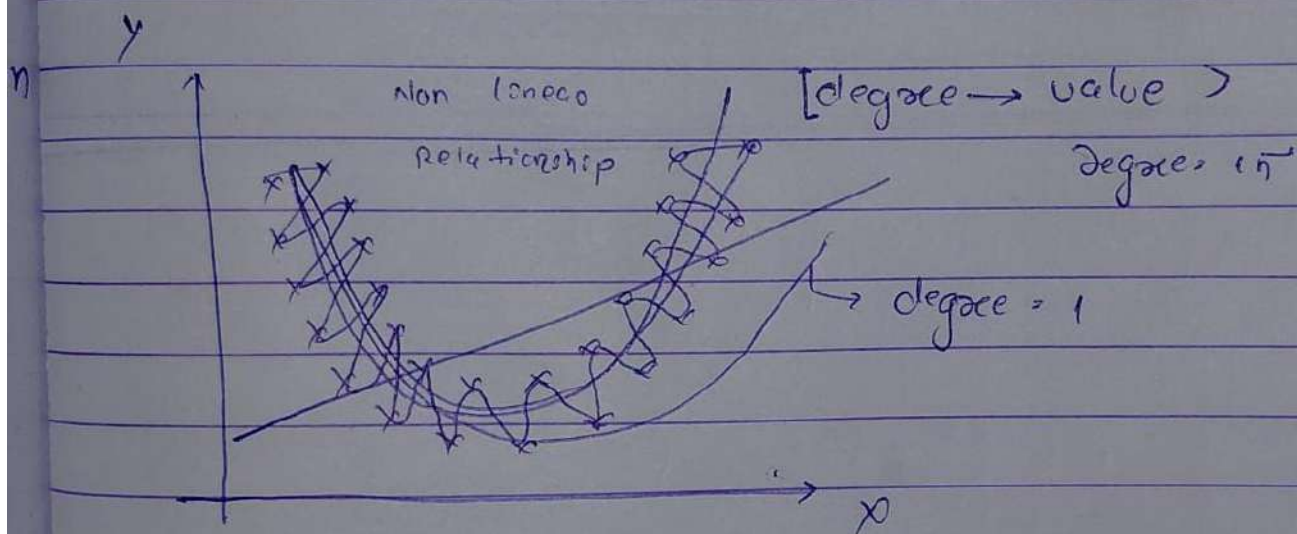
$$h_0(x) = \beta_0 * x^0 + \beta_1 * x^{(1)} \rightarrow \text{simple Linear Regression}$$

Polynomial
degree = 2

$$h_0(x) = \beta_0 * x^0 + \beta_1 * x^{(1)} + \beta_2 * x^{(2)}$$

Polynomial
degree = n

$$h_0(x) = \beta_0 * x^0 + \beta_1 * x^{(1)} + \beta_2 * x^{(2)} + \beta_3 * x^{(3)} + \dots + \beta_n * x^{(n)}$$



{ 2 independent feature }

degree = 1 $h_0(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$

degree = 2 $h_0(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1^2 + \beta_4 x_2^2$

↑